Peer Review

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1 Introduction

The topic is introduced clearly, with a good motivation on why it is relevant. The goal of the research is also clearly stated. Furthermore, there is a good justification on why an agent-based model is used for this research.

However, the introduction is missing some clarifications at certain parts. For example, the authors mention that Bautista extends bounded-confidence models, but they do not cite any papers for the bounded-confidence model. Furthermore, the authors do not explain what network topology entails. From the text I can infer that it is closely related to network structure, but it is not explained whether network topology is the same as network structure or if they are different concepts. On a similar note, the introduction mentions three different types of network topologies, but does not explain them. My suggestion would be to add a sentence stating that these topologies will be expanded upon later, or leave them out entirely until you can explain them. Similarly, the authors introduce the concept of stubborn agents as one of the three agent types without explaining their origin. Are they part of the extension made by Bautista, or were they already present in the original bounded-confidence model?

A small point of additional feedback is that the introduction mentions that 120 agents are simulated over 80 time steps, which are design choices. I would not include these in the introduction, instead only mention them in the methods.

Finally, the introduction contains a section about innovation. This section seems to repeat a lot of the points made in the topic introduction, and I would suggest merging the two sections.

2 Related Work

The related works section does a good job of explaining the research gap. It is clear that previous works have either looked at different topologies, or at the effects of stubborn and manipulative agents, but none have looked at both. Thus, the relevance of thee author's research is clear.

There are some small points that could be improved here. First, the authors again mention the HK, Deffuant-Weisbuch, and weighted HK models without explaining what they are. Similarly, without an explanation on what they are, the authors also mention the network topologies Erdős-Rényi random, Watts-Strogatz small-world, Barabási-Albert scale-free, and a fully connected network. As mentioned before, these concepts should be explained or a reference to a later section should be made (e.g. "These models will be explained in Section X"). Second, a very small point is that in the first paragraph it

is mentioned that a stubborn agent does change its opinion. This seems like a mistake, especially after reading Section 3.2.

3 Methods

3.1 Problem Specification

The problem specification is done clearly, with design choices being justified.

My only point of feedback would be to check spelling and grammar, as some sentences do not flow well and contain grammar mistakes. Finally, I would change "very interpretable" to "easily interpretable".

3.2 Agents

Two of the three types of agents are described clearly, and how each agent works is mostly clear.

In my opinion the normal and stubborn agents are described fully, but the manipulative agent could use some additional justification. First, the function $f(t)$ is poorly explained, and should be defined as a general function. Second, it is not well justified why a linear function is used. I understand that Bautista used a linear function as well, but it might be nice to include why they made that decision, as it's not immediately obvious.

3.3 Model Networks

The different model networks are explained well, and it is clear what each network attempts to represent.

In my opinion, this is the section that previous sections should have referred to when talking about these networks. That way it is clear to the reader that these network models are not assumed to be prior knowledge.

3.4 Implementation

The implementation is very well written, and the model is certainly reproducible from the descriptions. Furthermore, this section often refers back to previous sections, making it even more clear what each component represents.

3.5 Metrics

It is clear why each metric is being used and what their purpose is. However, I think it would be better if this section came after the experimental setup, since it references hyperparameters that have not been explained yet.

3.6 Experimental Setup

The experimental setup does explain the different experiments well, but it is missing a justification. It is not explained why 120 agents are used, and why only 20 of those are stubborn or manipulative. Furthermore, the simulation time is also not justified and thus feels arbitrary. Finally, I think this section would fit better before the implementation, as what is implemented is the experiments.

4 Results

The graphs and tables are clear, with a good description on the graph means and a nice visualization of the networks. Furthermore, the relevant hyperparameters used to obtain the graphs are also included.

However, the section overall could use some improvement. I assume that the random network is the baseline, but it is not explicitly said here. I would make that a bit clearer, maybe even create subsections for each network. As in the methods section, certain hyperparameters are not clearly justified. The authors mention $\epsilon = 0.1$, but a justification or reasoning behind this is missing. Furthermore, the results contain some possible explanations that should be in the discussion, in my opinion. Conversely, the expanded notation of the statistical significance should be in the results as well, not only in the discussion.

Finally, I believe that the authors should consider looking at Figure 2 again. It is quite different compared to the other two graphs, and the author's explanation is that this is likely by chance. In my opinion, this should be verified, and if it is indeed the case a different graph should be used. If after 10 runs this only happens once, it should be mentioned but not be presented as the main result of that experiment. In fact, the average over 10 runs might be more suitable to present, although I am not sure if that is possible.

5 Discussion

5.1 Interpreting the Results

The results are interpreted nicely, and plausible explanations for the results are given and justified by existing literature.

However, no additional explanation for the behavior in Figure 2 is given, which to me is the most interesting result. If it is indeed due to random chance as the authors claim, it would be interesting to try to discover how such behavior can emerge.

5.2 Limitations

The authors are clearly aware of their model's limitations, being very specific in how their model differs from real-life social networks. In particular, I think that the fact that network topology is intertwined with edge density and degree is very important, and it is good that the author's mention it.

6 Conclusion

The conclusion wraps up the paper nicely, and I believe the authors do not claim more than they showed. There are no suggestions for future work, which could be nice to add at the end.

7 Overall Thoughts

Overall, I think it is very clear why the research was performed. The research gap is clearly explained, and it is clear why an agent-based system works best here.

Some improvements could be made to the description of how the research was performed. The model description was clear, but certain design choices were not justified, certain concepts were not explained, and there was little explanation on how certain hyperparameter values (e.g. ϵ) were obtained.

Looking at the main takeaways of the research, it is clear what the authors want to convey. Additionally, the authors clearly explain how their results relate to existing literature.

Finally, I mostly agree with the approach of the authors. I agree that their research fills an existing gap, and I also agree with the methods they chose. The limitations are also clearly stated, and the conclusion is not too strong. However, I disagree with their display of results, as they use multiple seeds but only showcase the results of one seed.

In conclusion, I think the paper is nicely written, but could use some improvements in explaining concepts and justifying design choices. Also, the results section could be a bit more structured and better explained.